A. Assignment Information

Particular	Details
Department	Department of Computer Science and Engineering
Programme	B.E./B.Tech – Electronics and Communication Engineering
Course Code & Course Name	CSA0702 – Computer Networks
Academic Year / Batch	2024–2028
Faculty Name	Dr. SENTHAMILSELVAN R
Assignment Title	Design and Implementation of a VLAN-Based Enterprise LAN
Date of Issue	31-08-2026
Date of Submission	01-09-2026
Maximum Marks	100
Course Outcome(s) – CO	CO1, CO2, CO3, CO4, CO5
Bloom's Taxonomy Level	L6 – Design; L5 – Evaluate; L4 – Analyze; L3 – Configure; L2 – Explain
SDG Mapping	SDG 4 – Quality Education; SDG 9 – Industry, Innovation and Infrastructure; SDG 11 – Sustainable Cities and Communities
Industry / Societal Relevance	VLAN-based network segmentation is widely used in enterprises, educational institutions, and other organizations to improve network organization, security, traffic management, scalability, and reliability.

Course Outcome(s) – CO

CO1: Identify and explain fundamental networking concepts, network devices, topologies, protocols, and layered network architectures.

CO2: Configure IP addressing, subnetting, switching, and routing techniques for establishing computer networks using Cisco Packet Tracer.

CO3: Analyse network traffic, addressing information, VLAN communication, and connectivity using appropriate network simulation and packet-analysis tools.

CO4: Evaluate network performance, connectivity, segmentation, and security requirements for different enterprise networking scenarios.

CO5: Design and implement VLAN-based enterprise networks by creating separate VLANs for Administration, Faculty, Students, and Guests, configuring inter-VLAN communication, and verifying connectivity using Cisco Packet Tracer.

Bloom's Taxonomy Level

L6 – Design; L5 – Evaluate; L4 – Analyse; L3 – Configure; L2 – Explain.

SDG Mapping

- **SDG 4 – Quality Education:** Develops practical networking knowledge, digital skills, analytical abilities, and hands-on technical competency.
- **SDG 9 – Industry, Innovation and Infrastructure:** Promotes the design and implementation of reliable, secure, scalable, and innovative ICT infrastructure.
- **SDG 11 – Sustainable Cities and Communities:** Provides networking skills relevant to connected campuses, smart buildings, smart cities, and resilient digital services.

Industry / Societal Relevance

Modern organizations require secure and well-organized networks to support multiple departments and users. VLAN technology provides logical network segmentation, reduces unnecessary broadcast traffic, improves security, and simplifies network management. The proposed VLAN-based enterprise LAN is relevant to offices, educational institutions, industries, and smart infrastructure environments.

B. ASSIGNMENT PROBLEM / CHALLENGE

Design and implement a VLAN-based enterprise Local Area Network using Cisco Packet Tracer. The organization contains four departments:

1. Administration
2. Faculty
3. Students
4. Guests

Each department must be logically separated using a dedicated VLAN. Appropriate IP addressing, switch-port assignments, VLAN configuration, inter-VLAN communication, and connectivity testing must be implemented. The design should provide proper communication between authorized network segments while maintaining effective departmental segmentation.

C. PROBLEM STATEMENT

The objective of this assignment is to design and implement a VLAN-based enterprise LAN for an organization with four different user groups: Administration, Faculty, Students, and Guests. A traditional flat network places all devices in a single broadcast domain. This can result in excessive broadcast traffic, poor security, difficult network management, and unrestricted communication between unrelated departments. The proposed solution divides the enterprise network into four separate VLANs. Each VLAN represents a separate logical network. A Layer 3 device is used to provide controlled inter-VLAN communication.

The final network must:

- Create four VLANs.
- Assign switch ports to appropriate VLANs.
- Configure IP addresses for all devices.
- Configure trunk communication.
- Configure inter-VLAN routing.
- Verify connectivity using ping and switch/router commands.

- Analyze and troubleshoot possible configuration errors.

D. REQUIREMENTS AND CONSTRAINTS

1. Functional Requirements

The network shall:

- Support Administration, Faculty, Students, and Guests.
- Create separate VLANs for each department.
- Assign end devices to the correct VLAN.
- Provide IP addressing for each VLAN.
- Enable inter-VLAN routing.
- Verify successful communication.
- Maintain logical separation between departments.

2. Technical Requirements

- Cisco Packet Tracer must be used.
- A Cisco switch and router must be configured.
- IEEE 802.1Q trunking must be used between the switch and router.
- Unique IP subnets must be assigned to every VLAN.
- Router-on-a-Stick inter-VLAN routing must be implemented.

3. Constraints

- The network is implemented in a simulation environment.
- Available switch ports must be efficiently allocated.
- Each VLAN must use a different IP subnet.
- Incorrect trunking or VLAN assignment may cause communication failure.
- Guest network access should be logically isolated and controlled.

E. WORK

1. Problem Understanding and Formulation What

is the problem?

The problem is to design an enterprise network containing four different departments while preventing all devices from operating in one large broadcast domain.

Expected Outcomes

The expected outcomes are:

- Successful creation of four VLANs.
- Proper port-to-VLAN assignment.
- Correct IP addressing.
- Successful inter-VLAN routing.
- Successful connectivity testing.
- Identification and correction of configuration errors.

Available Information

The available information includes the four required departments and the requirement to use VLANs, IP addressing, inter-VLAN communication, and Cisco Packet Tracer.

Assumptions

The following assumptions are made:

- One switch is sufficient for the initial network design.
- One router is used for inter-VLAN routing.
- Each department has two PCs for testing.
- The router is connected to the switch using a trunk link.
- IPv4 addressing is used.

Constraints to be Satisfied

- Separate VLANs must be maintained.

- Every VLAN requires a unique subnet.
- Switch ports must be correctly assigned.
- The router must contain a sub interface for each VLAN.
- The trunk link must carry all required VLANs.

2. APPLICATION OF COURSE KNOWLEDGE

The following Computer Networks concepts are applied.

VLAN

A Virtual Local Area Network logically divides a physical network into multiple broadcast domains. Devices belonging to the same VLAN can communicate at Layer 2 even when separated from devices in other VLANs.

VLAN IDs

The following VLAN numbering scheme is used:

Department	**VLAN ID**	**VLAN Name**
Administration	10	ADMIN
Faculty	20	FACULTY
Students	30	STUDENTS
Guests	40	GUESTS

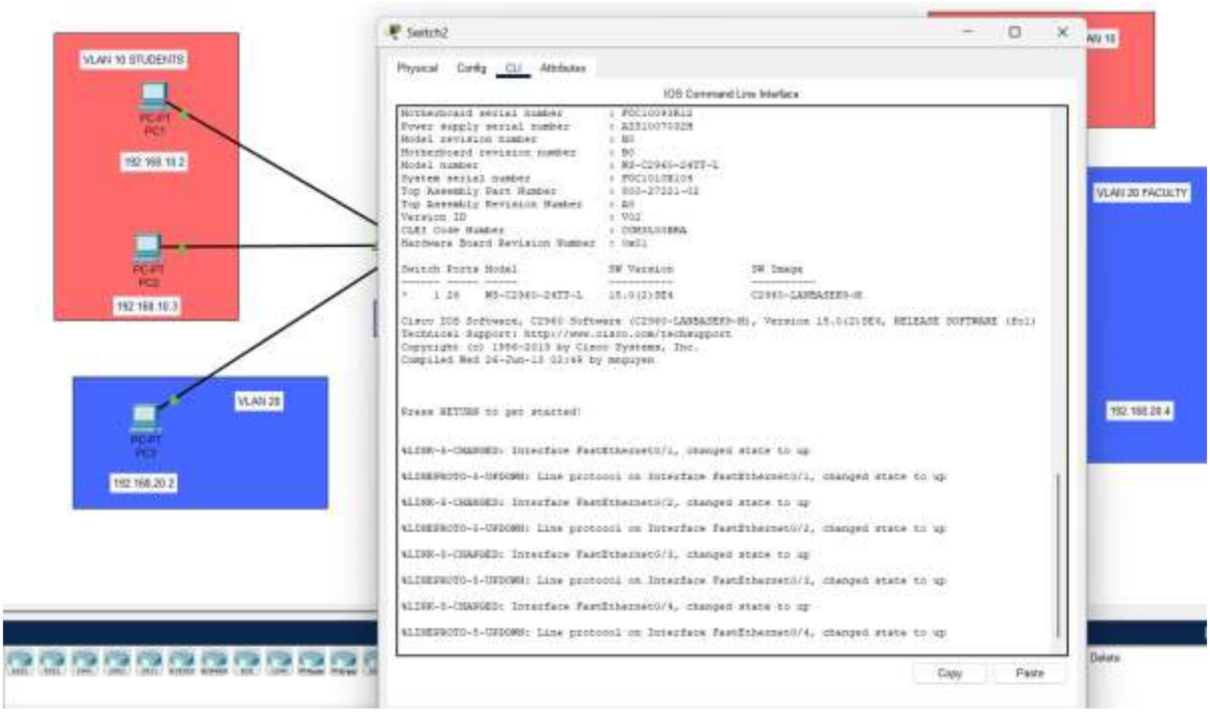
IP Addressing and Subnetting

Each VLAN receives a separate IPv4 subnet.

VLAN	**Network Address**	**Default Gateway**	**Subnet Mask**
VLAN 10	192.168.10.0/24	192.168.10.1	255.255.255.0
VLAN 20	192.168.20.0/24	192.168.20.1	255.255.255.0
VLAN 30	192.168.30.0/24	192.168.30.1	255.255.255.0

VLAN 40	192.168.40.0/24	192.168.40.1	255.255.255.0

Using separate subnets ensures that each VLAN represents a different logical IP network.



1. Fig:1- Troubleshooting/verification commands.

Inter-VLAN Routing

Devices in different VLANs cannot communicate directly through Layer 2 switching. A Layer 3 device is required. In this design, **Router-on-a-Stick** is used. The router interface is divided into multiple sub interfaces, with one sub interface assigned to each VLAN.

3. SOLUTION / DESIGN / METHODOLOGY

Proposed Network Topology

The proposed topology consists of:

- 1 Router
- 1 Cisco Switch
- 8 PCs
- 4 VLANs

Device Allocation

Device Group	Number of PCs	VLAN
Administration	2	VLAN 10
Faculty	2	VLAN 20
Students	2	VLAN 30
Guests	2	VLAN 40

Suggested Port Allocation

Switch Ports	Department	VLAN
Fa0/1 – Fa0/2	Administration	10
Fa0/3 – Fa0/4	Faculty	20
Fa0/5 – Fa0/6	Students	30
Fa0/7 – Fa0/8	Guests	40
Fa0/24	Router Trunk	All VLANs

Logical Topology

VLAN Configuration

The following VLANs are created on the switch:

enable configure terminal vlan 10

name ADMIN

vlan 20

name FACULTY

vlan 30

name STUDENTS

vlan 40

name GUESTS

end

Switch Port Assignment

enable configure

terminal interface range

fa0/1-2 switchport mode

access switchport access

vlan 10 interface range

fa0/3-4 switchport mode

access switchport access

vlan 20 interface range

fa0/5-6 switchport mode

access switchport access

vlan 30 interface range

fa0/7-8 switchport mode

access switchport access

vlan 40 end

Router-on-a-Stick Configuration

Assume the router is connected through interface GigabitEthernet0/0.

enable

configure terminal interface

gigabitEthernet0/0 no

shutdown

VLAN 10

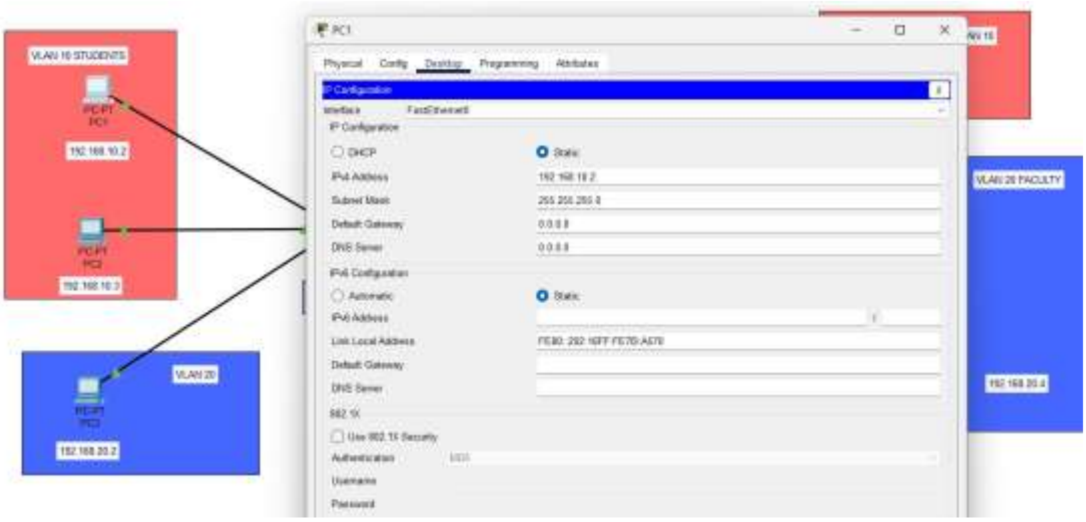


Fig: 2- IP configuration of PCs

interface gigabitEthernet0/0.10

encapsulation dot1Q 10

ip address 192.168.10.1 255.255.255.0

VLAN 20

interface gigabitEthernet0/0.20

encapsulation dot1Q 20 ip address

192.168.20.1 255.255.255.0 **VLAN 30**

interface gigabitEthernet0/0.30

encapsulation dot1Q 30 ip address

192.168.30.1 255.255.255.0

VLAN 40

interface gigabitEthernet0/0.40

encapsulation dot1Q 40

ip address 192.168.40.1 255.255.255.0

end write memory

Fig: 3- Switch port assignments

4. USE OF MODERN TOOLS

The assignment is implemented using **Cisco Packet Tracer**.

Cisco Packet Tracer is used to:

- Design the network topology.
- Add routers, switches, and PCs.
- Create VLANs.
- Configure switch ports.
- Configure router subinterfaces.
- Assign IP addresses.
- Test connectivity.
- Identify configuration errors.
- Simulate packet transmission.

Evidence of tool usage should include screenshots of:

1. Complete topology.

2. VLAN configuration.
3. Switch port assignments.
4. Trunk configuration.
5. Router subinterface configuration.
6. IP configuration of PCs.
7. Successful ping results.

The assignment format specifically expects evidence of modern-tool usage and outputs.

5. RESULTS AND VALIDATION

Expected Device IP Addresses

Device	IP Address	Gateway	VLAN
Admin-PC1	192.168.10.10	192.168.10.1	10
Admin-PC2	192.168.10.11	192.168.10.1	10
Faculty-PC1	192.168.20.10	192.168.20.1	20
Faculty-PC2	192.168.20.11	192.168.20.1	20
Student-PC1	192.168.30.10	192.168.30.1	30
Student-PC2	192.168.30.11	192.168.30.1	30
Guest-PC1	192.168.40.10	192.168.40.1	40
Guest-PC2	192.168.40.11	192.168.40.1	40

Validation Commands Verify

VLANs

show vlan brief

Expected result: VLANs 10, 20, 30, and 40 should be displayed with their assigned ports.

Verify Trunk show interfaces trunk

Expected result: Router sub interfaces should be configured with the correct IP addresses.

Connectivity Testing

From Admin-PC1: ping

192.168.10.11 ping

192.168.20.10 ping

192.168.30.10 ping

192.168.40.10

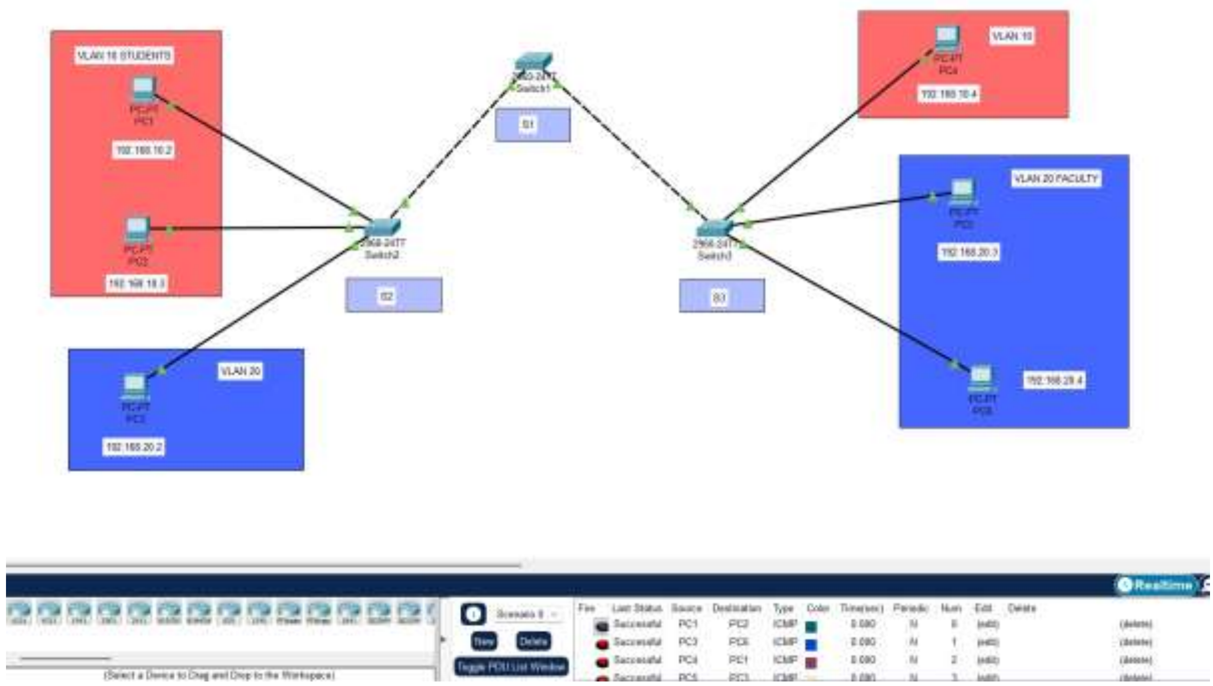


Fig:1- Successful replies confirm that VLAN configuration and inter-VLAN routing are functioning. The required assignment format expects validation through test results, screenshots, simulation outputs, and appropriate evidence that requirements are satisfied.

6. ANALYSIS AND ENGINEERING DECISION

Alternative Approaches

Two approaches were considered for inter-VLAN communication:

Approach 1: Router-on-a-Stick Advantages:

- Requires only one physical router interface.
- Lower hardware requirement.
- Suitable for small and medium enterprise networks.
- Easy to implement in Cisco Packet Tracer.

Limitation:

- All inter-VLAN traffic passes through one physical link. **Approach 2: Layer 3 Switch** Advantages:

- Faster inter-VLAN routing.
- Better scalability.
- Reduced routing bottleneck.

Limitations:

- Higher cost.
- More advanced configuration.

Final Decision

Router-on-a-Stick was selected because the assignment requires a manageable enterprise LAN simulation with limited devices. It provides an effective balance between simplicity, functionality, and implementation cost.

The VLAN design improves:

- Network organization.
- Broadcast traffic management.
- Departmental segmentation.
- Security.
- Scalability.

TROUBLESHOOTING AND ANALYSIS

Problem 1: PCs Cannot Communicate Within the Same VLAN Possible

Cause: Switch port assigned to the wrong VLAN.

Solution:

interface fa0/1 switchport

mode access switchport

access vlan 10

Problem 2: Different VLANs Cannot Communicate

Possible Cause: Router subinterfaces are missing or incorrectly configured.

Solution: Verify the following:

show ip interface brief

Configure the required subinterfaces with the correct VLAN IDs and IP addresses.

Problem 3: Trunk Link Not Working

Possible Cause: The switch port connected to the router is not configured as a trunk.

Solution:

interface fa0/24

switchport mode trunk

Verify using:

show interfaces trunk **Problem**

4: Ping Fails Possible

Causes:

- Incorrect IP address.
- Incorrect subnet mask.
- Incorrect default gateway.

- VLAN mismatch.
- Router interface shutdown.

Solution: Check PC IP configuration and execute:

show vlan brief show interfaces trunk show ip

interface brief

Then correct the identified configuration. Troubleshooting is especially important because the assessment rubric allocates significant marks to independently identifying and resolving issues.

7. BROADER CONSIDERATIONS

Security

VLAN segmentation provides logical separation between departments. Sensitive Administration traffic can be separated from Student and Guest traffic.

Scalability

Additional departments can be added by creating new VLANs and allocating new IP subnets.

Cost

The Router-on-a-Stick design reduces hardware requirements because multiple VLANs use one physical router interface.

Sustainability

Efficient network design reduces unnecessary broadcast traffic and supports reliable digital infrastructure.

Professional Responsibility

Network engineers must configure secure, reliable, and maintainable networks. Proper VLAN design reduces the possibility of unauthorized access and network misconfiguration. The common format specifically calls for consideration of broader impacts such as sustainability, society, safety, ethics, economics, and professional responsibility where relevant.

8. CONCLUSION

A VLAN-based enterprise LAN was successfully designed for Administration, Faculty, Students, and Guests. Four separate VLANs were created to logically divide the network into independent broadcast domains. The network was configured using Cisco Packet Tracer with appropriate VLAN IDs, switch port assignments, IPv4 addressing, trunking, and Router-on-aStick inter-VLAN routing. The results demonstrate that VLAN technology improves network organization, reduces unnecessary broadcast traffic, and provides better security through logical segmentation. Connectivity was validated using VLAN verification commands, interface checks, and ping testing. The design satisfies the primary assignment requirements. A future improvement would be to use a Layer 3 switch, access control lists, DHCP, redundant switches, and additional security policies for a larger enterprise network.

9. REFLECTION

This assignment provided practical experience beyond theoretical understanding of VLANs and IP addressing. I learned how a physical network can be logically divided into multiple independent networks using VLAN technology. I gained practical knowledge of creating VLANs, assigning switch ports, configuring trunk links, implementing Router-on-aStick inter-VLAN routing, and troubleshooting connectivity problems. I also understood the importance of correct IP addressing and default gateway configuration. If additional time and resources were available, I would improve the network by implementing a Layer 3 switch, Access Control Lists for restricting Guest access, DHCP for automatic IP allocation, redundant network devices, and stronger enterprise security mechanisms. This assignment also demonstrates the relevance of **SDG 9**, as reliable and scalable network infrastructure is essential for modern organizations, industries, educational institutions, and digital services.

10. REFERENCES

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